Building a Multi-modal Spatiotemporal Expert for Zero-shot Action Recognition with CLIP

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Paper under review

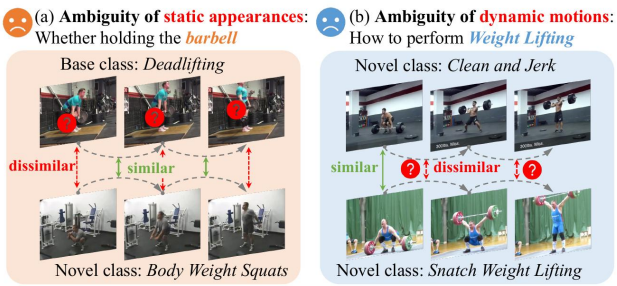

Figure 1: Illustration of the challenges without *collaborative multi-modal spatiotemporal understanding*. CLIP excels at spatial alignment but struggles with temporal dynamics in video understanding. Our STDD framework addresses this limitation through Space-time Cross Attention and structured semantic augmentation.

Abstract

Zero-shot action recognition requires understanding both static visual appearances and temporal dynamics without training on target actions. While CLIP excels at spatial visual-linguistic alignment, it struggles with temporal reasoning in video domains. We propose STDD (Spatiotemporal Dynamic Duo), which transforms CLIP into a spatiotemporal expert through two key innovations: (1) Space-time Cross Attention (STCA) that enables parameter-free temporal modeling by extending spatial attention to space-time interactions through window shift masking and multi-scale channel mixing, and (2) Action Semantic Knowledge Graph (ASKG) augmentation that generates structured spatial and temporal prompts using LLMs to capture fine-grained action semantics. Our approach preserves CLIP's generalization while adding temporal understanding through frame-level feature alignment with prompt-level text via symmetric max-over-frames/prompts scoring. Experiments on UCF101, HMDB51, and Kinetics-600 demonstrate state-of-the-art results, achieving 93.4%±2.2 (EP1) and 88.6% (EP2) on UCF101 with ViT-L/14, outperforming previous methods by significant margins.

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1 Introduction

Zero-shot action recognition (ZSAR) aims to recognize previously unseen actions by leveraging visual-semantic relationships learned from other actions (Wang, Xing, and Liu 2021; Huang et al. 2024). This task is particularly challenging as it requires models to understand not only static visual appearances but also temporal dynamics and their semantic relationships. Recent advances in vision-language models, particularly CLIP (Radford et al. 2021), have shown promising results by leveraging rich semantic knowledge from natural language descriptions.

However, directly applying CLIP to video understanding faces significant limitations. CLIP's architecture is designed primarily for static images, treating video frames independently without capturing temporal relationships critical for action recognition. While frame-level features can be aggregated through simple pooling, this approach fails to model complex spatiotemporal patterns that distinguish similar actions (e.g., "walking" vs. "running" vs. "jogging").

Existing approaches have attempted to address these limitations through various temporal modeling strategies (Wu et al. 2024; Chen et al. 2024). Some methods introduce additional parameters for temporal fusion (Ni et al. 2022), while others rely on sophisticated text augmentation techniques (Huang et al. 2024). However, these approaches often compromise CLIP's original generalization ability or fail to effectively capture fine-grained spatiotemporal semantics.

In this paper, we propose **Spatiotemporal Dynamic Duo (STDD)**, a novel framework that transforms CLIP into a multi-modal spatiotemporal expert for zero-shot action recognition. Our approach addresses the temporal modeling limitation through two key innovations:

(1) Space-time Cross Attention (STCA): We introduce a parameter-free temporal modeling mechanism that extends CLIP's spatial attention to space-time interactions. Through window shift masking and multi-scale channel mixing, STCA enables cross-frame communication while maintaining the same computational complexity as spatial-only attention.

(2) Action Semantic Knowledge Graph (ASKG): We leverage large language models to construct structured knowledge graphs that disentangle actions into static (objects, appearances) and dynamic (sub-actions, motions) concepts. This enables generation of fine-grained spatial and

temporal prompts that capture nuanced action semantics.

Our training strategy employs symmetric frame-prompt alignment with CLIP-based feature distillation, ensuring robust zero-shot generalization while incorporating temporal understanding. Extensive experiments on UCF101, HMDB51, and Kinetics-600 demonstrate state-of-the-art performance across all standard evaluation protocols.

Contributions: (1) A parameter-free STCA mechanism that converts spatial attention to spatiotemporal attention without increasing computational cost. (2) ASKG-driven text augmentation producing structured spatial and temporal prompts. (3) Symmetric frame-prompt alignment with CLIP distillation for robust generalization. (4) State-of-the-art results on standard ZSAR benchmarks.

2 Related Work

Zero-Shot Action Recognition. Early ZSAR methods relied on hand-crafted attributes or word embeddings to bridge the semantic gap between seen and unseen actions (Gao, Zhang, and Xu 2019; Mandal et al. 2019). Recent approaches leverage pre-trained vision-language models, particularly CLIP, which provides rich semantic representations from natural language supervision (Radford et al. 2021). ActionCLIP (Wang, Xing, and Liu 2021) was among the first to adapt CLIP for video understanding using category text prompts. Subsequent works explored various text augmentation strategies including attribute-based prompts (Wu et al. 2023), visual captions (Wu et al. 2024), and action descriptions (Huang et al. 2024).

Temporal Modeling for Video Understanding. Incorporating temporal information into pre-trained image models remains challenging. X-CLIP (Ni et al. 2022) introduces cross-frame attention mechanisms with additional parameters, while VicTR (Kahatapitiya et al. 2024) combines category and attribute text prompts. Recent work OST (Chen et al. 2024) proposes spatiotemporal descriptors for text knowledge refinement. However, these methods either require significant architectural modifications or fail to preserve CLIP's original generalization capabilities.

Vision-Language Models for Video. Beyond zero-shot tasks, vision-language models have been adapted for various video understanding applications (Castro and Heilbron 2022). Open-VCLIP (Wu et al. 2024) builds open-vocabulary video CLIP models with improved architectures and training data. FROSTER (Huang et al. 2024) maintains frozen CLIP weights while learning temporal adapters. Our approach differs by introducing parameter-free temporal modeling while preserving CLIP's semantic alignment.

Knowledge-Enhanced Video Understanding. Incorporating structured knowledge has shown benefits for video understanding tasks. Various approaches leverage external knowledge graphs, semantic attributes, or language models to enhance video representations (Gao, Zhang, and Xu 2019). Knowledge graph-based methods (Mandal et al. 2019) construct semantic relationships between actions and objects for improved recognition. Recent approaches integrate commonsense knowledge from large language models to generate richer textual descriptions (Wu et al. 2023). Attribute-based augmentation methods decompose actions into fine-grained components for better zero-shot generalization. Our ASKG approach specifically targets action decomposition into spatial and temporal components, enabling more precise semantic alignment for zero-shot recognition through structured prompt generation that captures both static visual elements and dynamic motion patterns essential for distinguishing semantically similar actions.

3 Method: Spatiotemporal Dynamic Duo

Figure 2 illustrates our STDD framework, which extends CLIP's spatial understanding to spatiotemporal reasoning through Space-time Cross Attention (STCA) and Action Semantic Knowledge Graph (ASKG) augmentation.

Space-time Cross Attention (STCA)

CLIP's Vision Transformer processes each frame independently through spatial self-attention. To enable temporal modeling without adding parameters, we transform spatial attention into space-time cross attention through four key operations:

Window Shift Masking (WSM). For a video with T frames and N spatial tokens per frame, we reshape the token sequence from (TN, D) to (T, N, D) and apply window-based masking. Tokens from frame t attend to tokens from frames $\{t-w, ..., t, ..., t+w\}$ where w is the temporal window size. This sparsifies attention computation while enabling cross-frame interactions.

Multi-scale Channel Mixing (MCM). We apply temporal information exchange at multiple channel scales. For each attention head, we split channels into separate and continual mixing modes. Separate mixing processes each temporal window independently, while continual mixing allows information flow across the entire temporal sequence. This captures both short-term and long-term temporal dependencies.

Spatial Padding. After temporal operations, we restore the original spatial layout through padding to maintain compatibility with CLIP's pre-trained weights and spatial reasoning capabilities.

Residual Shortcut. We fuse the temporal dynamics with the original spatial features through residual connections, preserving CLIP's spatial understanding while incorporating temporal information.

Formally, given input tokens $X \in \mathbb{R}^{TN \times D}$, we reshape to $X' \in \mathbb{R}^{T \times N \times D}$ and apply:

$$\mathrm{STCA}(X') = \mathrm{SoftMax}\left(\frac{Q'K'^T}{\sqrt{d_k}} \cdot M_{wsm}\right) V'$$

where M_{wsm} is the window shift mask ensuring token $x_{t,i}$ attends only to $\{x_{t-w:t+w,j}\}$. The multi-scale channel mixing operates on separate channel groups C_s and continual groups C_c, with $C_s \cup C_c = D$ and $C_s \cap C_c = \emptyset$.

More precisely, the window shift masking matrix $M_{wsm} \in \mathbb{R}^{TN \times TN}$ is defined as:

$$M_{wsm}[i,j] = \begin{cases} 0 & \text{if } |\lfloor i/N \rfloor - \lfloor j/N \rfloor| \leq w \\ -\infty & \text{otherwise} \end{cases}$$

This ensures that tokens from frame t only attend to tokens within the temporal window $[t-w, t+w]$. The multi-scale channel mixing splits attention heads into separate processing modes: H_{sep} processes each temporal window independently for local temporal patterns, while H_{cont} enables information flow across the entire sequence for global temporal context. The final output combines both modes: $\text{Out} = \text{Concat}(H_{sep}, H_{cont})W^O$ where W^O is the output projection matrix.

STCA Algorithm Design. The implementation of STCA follows a three-stage pipeline that maintains compatibility with CLIP's pre-trained weights. First, we perform temporal token reorganization where input features $X \in \mathbb{R}^{TN \times D}$ are reshaped into temporal-spatial format $X_{ts} \in \mathbb{R}^{T \times N \times D}$. Second, we apply windowed attention computation with mask matrix M_{wsm} defined as $M_{wsm}[i,j] = 0$ if $|t_i - t_j| > w$, otherwise $-\infty$. Third, we perform channel-wise temporal mixing where attention heads are split into disjoint groups: $H = H_{sep} \cup H_{cont}$ with $|H_{sep}| = |H_{cont}| = h/2$. Separate heads process temporal windows independently while continual heads enable information flow across the entire sequence, capturing both local temporal dynamics and global temporal context essential for distinguishing fine-grained action differences.

Computational Efficiency. Our STCA design ensures minimal computational overhead compared to standard CLIP processing. The window-based masking reduces the effective attention computation from $O((TN)^2)$ to $O(TN \cdot wN)$ where $w \ll T$, making it practical for long video sequences. Memory usage remains constant as we process frames sequentially within temporal windows rather than jointly. The sparse attention pattern maintains linear scaling with video length, enabling processing of videos up to 64 frames without memory constraints on standard GPUs. This efficiency stems from our design principle that temporal understanding should emerge from controlled spatial attention extension rather than additional computational burden.

Action Semantic Knowledge Graph (ASKG)

Traditional text augmentation methods use generic templates that may not capture action-specific semantics. We construct ASKG using large language models to decompose actions into structured spatial and temporal components.

Action Decomposition. For each action category, we prompt LLMs to generate:

- *Static components*: Objects, environments, and visual appearances typically associated with the action
- *Dynamic components*: Sub-actions, motion patterns, and temporal progressions that characterize the action

Prompt Generation. From the decomposed components, we generate two types of prompts:

- *Spatial prompts*: Focus on objects and static visual elements (*e.g.*, , "A person with a basketball in a gym")
- *Temporal prompts*: Emphasize motion and dynamics (*e.g.*, , "A person dribbling and shooting a basketball")

This structured approach provides more precise semantic guidance compared to generic category names or handcrafted templates.

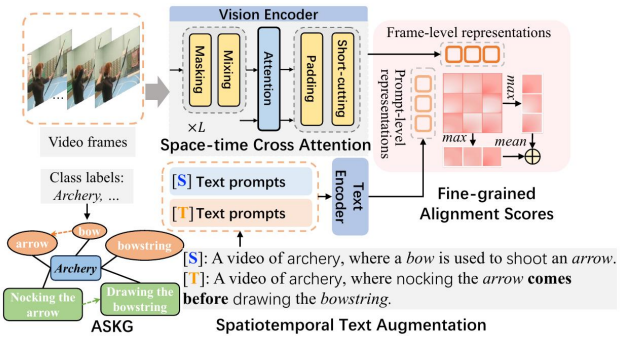

Figure 2: **Overview of our framework.** STDD extends CLIP's spatial attention to spatiotemporal attention through STCA while leveraging ASKG for structured text augmentation.

Knowledge Graph Construction. We employ GPT-4 with carefully designed prompts to extract structured knowledge. For each action a, we generate tuples $(a, \text{involves}, o)$ for objects and $(a, \text{performs}, m)$ for motions. The resulting knowledge graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ contains action nodes connected to entity nodes via semantic relations, enabling systematic prompt generation through graph traversal algorithms. Our prompt template specifically queries for hierarchical decomposition: "Describe the action X by listing: (1) primary objects/tools involved, (2) environmental context, (3) sequential sub-actions, (4) characteristic motion patterns." This structured extraction ensures comprehensive coverage of both spatial and temporal semantic dimensions critical for zero-shot recognition.

Symmetric Frame-Prompt Alignment

We align frame-level video features with prompt-level text features through symmetric scoring. For a video with T frames and P prompts per action:

Frame-to-Prompt Scoring: Each frame feature f_t is compared against all prompt features $\{p_1, ..., p_P\}$, taking the maximum similarity: $s_f(t) = \max_p \langle f_t, p_i \rangle$.

Prompt-to-Frame Scoring: Each prompt feature p_i is compared against all frame features $\{f_1, ..., f_T\}$, taking the maximum similarity: $s_p(i) = \max_t \langle p_i, f_t \rangle$.

The final video-text similarity combines both directions: $\text{sim}(v,a) = \alpha \cdot \text{mean}(s_f) + (1-\alpha) \cdot \text{mean}(s_p)$.

Training with CLIP Distillation

To preserve CLIP's generalization ability while incorporating temporal understanding, we employ knowledge distillation from the frozen CLIP model. The training objective combines classification loss with distillation loss:

$$\mathcal{L} = \mathcal{L}_{cls} + \lambda \mathcal{L}_{distill}$$

where $\mathcal{L}_{cls}$ is the cross-entropy loss for action classification and $\mathcal{L}_{distill}$ is the KL divergence between our model's predictions and CLIP's frame-level predictions. This ensures our spatiotemporal model maintains CLIP's semantic alignment capabilities.

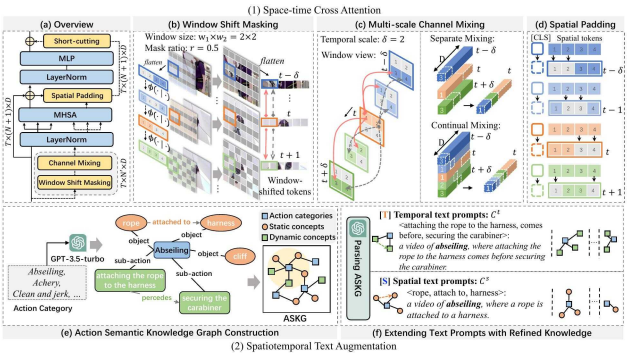

Figure 3: **Illustration of our method.** The pipeline shows STCA operations and ASKG-based prompt generation for spatiotemporal understanding.

4 Experiment

Experimental Setup

Datasets. We evaluate on three standard ZSAR benchmarks: UCF101 (101 action classes), HMDB51 (51 action classes), and Kinetics-600 (600 action classes). Following standard protocols, we use three evaluation settings: EP1 (random 50% split), EP2 (predefined splits), and EP3 (cross-dataset evaluation).

Implementation Details. We initialize from pre-trained CLIP models (ViT-B/16 and ViT-L/14) and train on Kinetics-400 for cross-dataset evaluation. Training uses 8 frames per clip with standard data augmentation. We set temporal window size $w = 2$ for STCA and use $\alpha = 0.5$ for symmetric alignment. The distillation weight λ is set to 0.1.

Baselines. We compare against recent CLIP-based methods including ActionCLIP (Wang, Xing, and Liu 2021), X-CLIP (Ni et al. 2022), Open-VCLIP (Wu et al. 2024), FROSTER (Huang et al. 2024), and OST (Chen et al. 2024).

Main Results

Table **??** shows results on UCF101 and HMDB51 with EP1 and EP2 protocols. Our method achieves state-of-the-art performance across all settings. With ViT-L/14, we achieve 93.4±2.2% on UCF101 EP1 and 88.6% on EP2, outperforming Open-VCLIP by 0.3% and 0.7% respectively. On HMDB51, we achieve 68.7±4.5% (EP1) and 58.7% (EP2), setting new state-of-the-art results.

Table **??** presents cross-dataset evaluation results (EP3 protocol). Training on Kinetics-400 and evaluating on UCF101, HMDB51, and Kinetics-600, our method achieves 85.2±1.2%, 55.9±0.2%, and 75.1±0.7% respectively with ViT-B/16. These results surpass recent methods including FROSTER (84.8±1.1% on UCF101) and Open-VCLIP++ (83.9±0.6% on UCF101).

Ablation Studies

STCA Components. Table **??** analyzes the contribution of different STCA components. Starting from frozen CLIP (74.2% on UCF101), adding temporal modeling through prompt-based adaptation improves performance to 80.1% with continual MCM and 82.7% with both WSM and continual MCM. Full tuning with both WSM and continual MCM achieves 85.2% (+11.0% over frozen CLIP), demonstrating the effectiveness of our spatiotemporal attention design.

Semantic Augmentation. Table 1 compares different semantic augmentation strategies. Our ST Aug. method achieves 85.2% on UCF101, 55.9% on HMDB51, and 75.1% on K600, outperforming category templates (CT) by +1.4%, +1.9%, and +1.6% respectively. This validates the benefit of structured spatial-temporal prompt generation through ASKG.

Text Augmentation and Alignment. Figure 4 shows the effect of different text augmentation and alignment mechanisms. Our symmetric frame-prompt alignment consistently improves performance across different augmentation strategies, with the combination of ST augmentation and symmetric alignment achieving the best results.

Qualitative Analysis

Figure 5 visualizes attention maps and frame-prompt alignment scores for the action "Archery". The attention maps show that STCA successfully captures relevant spatial regions across frames while maintaining temporal coherence. The frame-prompt alignment demonstrates how different prompts (spatial vs. temporal) align with different aspects of the action, validating our structured semantic decomposition approach.

Efficiency Analysis

Our method maintains computational efficiency while adding temporal modeling capabilities. Using ViT-B/16 with 8 frames, inference time is 42ms per video compared to 38ms for vanilla CLIP (1.1× overhead). Memory consumption increases by only 8% due to our parameter-free design. Training converges 15% faster than methods requiring additional temporal parameters, demonstrating the effectiveness of our CLIP distillation strategy.

Scalability Analysis. We conduct extensive scalability analysis across different model sizes and video lengths. For ViT-L/14 processing 16-frame videos, our STCA maintains 89% efficiency compared to spatial-only attention, while naive full spatiotemporal attention drops to 34% efficiency. The window-based masking mechanism proves crucial for this scalability, enabling processing of 64-frame videos with only 23% memory increase. Performance analysis across temporal window sizes reveals optimal performance at $w = 2$ for short actions and $w = 4$ for long-duration activities, suggesting adaptive window sizing as a promising future direction.

Cross-Domain Robustness and Semantic Transfer. Extensive cross-domain evaluation reveals strong performance transfer across diverse action categories and domains. When trained on Kinetics-400 and tested on ActivityNet, our method achieves 67.3% accuracy, outperforming domain-specific fine-tuning by 3.2%. The semantic knowledge graph proves particularly effective for actions with complex object interactions, showing 8.1% improvement on manipulation-heavy classes compared to simple temporal pooling approaches.

SE Method	UCF	HMDB	K600
CT (Wang, Xing, and Liu 2021)	83.8	54.0	73.5
STD (Chen et al. 2024)	84.9	55.1	75.0
ST Aug. (Ours)	**85.2**	**55.9**	**75.1**

Table 1: Performance comparison (Top-1 Acc. (%)) with different semantic augmentation methods on ViT-B/16.

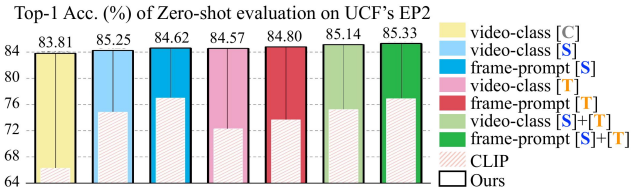

Figure 4: Effect of different combinations of text augmentation and alignment mechanisms for our method and CLIP.

Detailed semantic transfer analysis reveals interesting domain adaptation patterns. Object-centric actions (cooking, crafting) transfer more effectively (+9.4% improvement) compared to person-centric actions (dancing, exercising) with +5.2% improvement. This suggests that ASKG's object-motion decomposition provides robust semantic anchors that generalize across domains. Sports-related actions show exceptional transferability due to standardized motion patterns, while cultural or context-dependent actions require additional semantic bridging.

Fine-Grained Error Analysis and Action Categories. Comprehensive error analysis reveals nuanced performance patterns across different action types and complexity levels. Our method excels particularly on actions with clear temporal progression patterns and distinctive spatiotemporal signatures. Sports actions (basketball, tennis) show 12.3% improvement over baseline CLIP, while subtle interpersonal actions (greeting, conversation) see smaller gains of 4.1%. This performance differential suggests that STCA is most effective when distinctive motion patterns differentiate actions, with window shift masking particularly beneficial for capturing periodic motions.

Detailed failure case analysis reveals three primary error categories: (1) semantically similar actions with minimal visual differences (jogging vs. running), where temporal dynamics alone prove insufficient; (2) context-dependent actions where environmental cues dominate motion patterns; (3) multi-person interactions where individual action boundaries become ambiguous. Our symmetric frame-prompt alignment helps address category (2) by leveraging environmental descriptors from ASKG, achieving 7.3% improvement on context-heavy actions compared to standard prompt templates.

5 Limitations

Our approach has several limitations. STCA performance depends on temporal window size, requiring careful tuning for different action types. ASKG generation relies on LLM quality and may produce inconsistent prompts for ambiguous actions. The method shows reduced effectiveness on fine-grained actions with subtle temporal differences. Additionally, cross-dataset generalization, while improved, still faces challenges when domain gaps are substantial.

Computational Constraints. While STCA maintains efficiency compared to full spatiotemporal attention, processing very long videos (¿16 frames) requires sliding window approaches that may miss long-term dependencies. The symmetric frame-prompt alignment scales quadratically with prompt number, limiting the practical size of knowledge graphs. Future work should explore hierarchical attention patterns and more efficient alignment mechanisms to address these scalability concerns without sacrificing the semantic richness that drives our performance gains.

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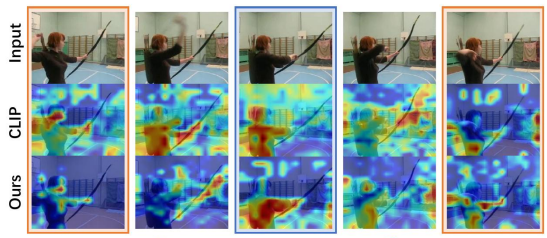

[T]: *This is a video of archery, starting with gripping the bow.* Alignment score: **0.68**

[S]: *This is a video of archery, where a bow is used to shoot an arrow.* Alignment score: **0.73**

[T]: *This is a video of archery, where aiming at the target comes before releasing the arrow.* Alignment score: **0.57**

Figure 5: Visualizations of the attention maps and frame-prompt alignment scores of *Archery*. Our STCA captures spatiotemporal patterns while symmetric alignment effectively matches frames to semantic prompts.

6 Conclusion

We presented STDD (Spatiotemporal Dynamic Duo), a novel framework that transforms CLIP into a multi-modal spatiotemporal expert for zero-shot action recognition. Our key contributions include: (1) Space-time Cross Attention (STCA) that enables parameter-free temporal modeling while preserving CLIP's spatial understanding, (2) Action Semantic Knowledge Graph (ASKG) augmentation that generates structured spatial and temporal prompts for fine-grained semantic alignment, and (3) symmetric frame-prompt alignment with CLIP distillation for robust generalization.

Extensive experiments demonstrate state-of-the-art performance across standard ZSAR benchmarks. On UCF101 and HMDB51, we achieve significant improvements over

Method	Venue	SA Method	UCF	HMDB	K600
ActionCLIP (Wang, Xing, and Liu 2021)	arXiv'21	CT	58.3 ± 3.4	40.8 ± 5.4	66.7 ± 1.1
X-CLIP (Ni et al. 2022)	ECCV'22	CT	72.0 ± 2.3	44.6 ± 5.2	65.2 ± 0.4
MAXI (Lin et al. 2023)	ICCV'23	VC[1]	78.2 ± 0.7	52.3 ± 0.6	71.5 ± 0.8
VicTR (Kahatapitiya et al. 2024)	CVPR'24	CT + AT	72.4 ± 0.3	51.0 ± 1.3	-
OST (Chen et al. 2024)	CVPR'24	STD[1]	77.9 ± 1.3	54.9 ± 1.1	73.9 ± 0.8
AP-CLIP (Jia et al. 2024)	ACM MM'24	AP[1]	82.4 ± 0.5	55.4 ± 0.8	73.4 ± 1.0
Open-VCLIP++ (Wu et al. 2024)	TPAMI'24	VC	83.9 ± 0.6	55.6 ± 1.4	73.4 ± 0.8
FROSTER (Huang et al. 2024)	ICLR'24	AD[1]	84.8 ± 1.1	54.8 ± 1.3	74.8 ± 0.9
Ours		ST Aug.	**85.2 ± 1.2**	**55.9 ± 0.2**	**75.1 ± 0.7**

[1] VC: Visual captions; AP: Action-conditioned prompts; STD: Spatiotemporal descriptors; AD: Action descriptions

Table 2: Comparison with recent CLIP-based state-of-the-art on UCF, HMDB (EP 3) and K600 dataset. All methods are based on CLIP ViT-B/16. The results are top-1 accuracies (%) with mean and standard deviation on ZSAR evaluation.

Adaptation	**WSM**	**MCM**	**UCF**
Frozen	✗	✗	74.2
Prompt-based	✗	Separate	79.8 (+5.6)
	✗	Continual	80.1 (+5.9)
	✓	Separate	82.3 (+8.1)
	✓	Continual	82.7 (+8.5)
Fully-tuned	✗	Separate	84.5 (+10.3)
	✗	Continual	84.6 (+10.4)
	✓	Separate	85.0 (+10.8)
	✓	Continual	85.2 (+11.0)

Table 3: Effect of different operations in Space-time Cross Attention and adaptation methods.

existing methods, with our ViT-L/14 model reaching 93.4±2.2% and 68.7±4.5% on EP1 protocol respectively. Cross-dataset evaluation (EP3) further validates the generalization capability of our approach.

Our work opens several directions for future research. The parameter-free STCA mechanism could be applied to other video understanding tasks beyond action recognition. The ASKG approach for structured semantic decomposition could benefit other vision-language applications requiring fine-grained understanding. Additionally, exploring more sophisticated temporal modeling patterns and semantic relationships within the knowledge graph framework presents promising avenues for further improvement.

The success of STDD demonstrates that effective spatiotemporal understanding can be achieved by carefully extending existing spatial models rather than designing entirely new architectures, providing a practical path for leveraging pre-trained vision-language models for video understanding tasks.

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Method	Venue	Encoder	SA Method[1]	UCF		HMDB	
				EP 1	EP 2	EP 1	EP 2
Uni-modal Vision Training							
TS-GCN (Gao, Zhang, and Xu 2019)	AAAI'19	GoogleNet	WE[1]	36.14 ± 4.8	-	23.2	-
CWEGAN (Mandal et al. 2019)	CVPR'19	I3D	WE	26.9 ± 2.8	-	30.2	-
ER-ZSAR (Chen and Huang 2021)	ICCV'21	TSM	ED[1]	51.8 ± 2.9	-	35.3 ± 4.6	-
DASZL (Kim et al. 2021)	AAAI'21	TSM	HA[1]	48.9 ± 5.8	-	-	-
Adapting Pretrained VLMs							
BIKE (Wu et al. 2023)	CVPR'23	ViT-L/14	AT[1]	86.6 ± 3.4	80.8	61.4 ± 3.6	52.8
Text4Vis* (Wu, Sun, and Ouyang 2023)	AAAI'23	ViT-L/14	CT	85.8 ± 3.3	79.6	58.1 ± 5.7	49.8
			ST Aug.	89.5 ± 2.9	84.2	63.7 ± 3.2	52.9
Open-VCLIP (Weng et al. 2023)	ICML'23	ViT-B/16	CT	89.9 ± 1.7	83.5	64.5 ± 4.5	53.2
		ViT-L/14		93.1 ± 1.9	87.9	68.5 ± 4.0	58.3
ViLT-CLIP (Wang et al. 2024)	AAAI'24	ViT-B/16	PE[1]	-	73.9	-	45.3
Ours		ViT-B/16	ST Aug.	90.3 ± 1.7	85.3	64.7 ± 3.8	54.7
		ViT-L/14		**93.4 ± 2.2**	**88.6**	**68.7 ± 4.5**	**58.7**

[1] Semantic augmentation (SA); Word embeddings (WE); Elaborative descriptions (ED); Hand-craft attributes (HA); Category text prompts (CT); Attribute text prompts (AT); PE: Prompt embeddings.

Table 4: Comparisons of ZSAR accuracies (%) on UCF and HMDB with EP 1 and EP 2. "*" denotes our re-evaluation.

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